

Question 1.

```
SELECT DISTINCT Department
FROM Student-table;
```

Output

Department.
IT
HR
Finance

Question 2.

```
SELECT Department,
       AVG (Age) As Average - Age
FROM Student-table
GROUP BY Department;
```

Output

Department	Average - Age
IT	20,5
HR	22
Finance	23

Question 3.

```
SELECT      Department,
            COUNT(student-ID) AS Number of learners
GROUP BY    Department
HAVING      number of learners > 1;
```

Output.

Department	Number of learners.
IT	2
HR	2

Question 4.

```
SELECT      Student-ID,
            Name,
            Age,
            department
FROM        student-table
WHERE       AGE BETWEEN 21 AND 23.
```

Output.

Student-ID	Name	Age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

Question 5

SELECT student-ID,
 Name,
 Age,
 Department
FROM student-table.
WHERE department IN ('IT', 'HR') AND AGE > 21;

Output.

Student-ID	Name	Age	Department
2	Bob	22	HR
5	Eve	22	HR

Question 6.

~~SELECT department,
 sum(credits)
FROM courses-table~~

Question 6

SELECT department,
 sum(credits) As Total-credits
FROM courses-table
GROUP BY Department
HAVING total-credits > 5;

Output.

Department	total-credits
IT	11

Question 7.

```
SELECT *  
FROM courses-table  
WHERE credits < > 4;
```

Output

Course-ID	Course-name	department	Credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

Question 8

```
SELECT Course-ID,  
        Course-name,  
        Credits  
FROM course-table  
ORDER BY Credits DESC  
LIMIT 3
```

Output

Course-ID	Course-name	Credits
102	Python	4
103	Data Science	4
105	Statistics	3

Question 9.

SELECT MAX (Grade) As max-grade,
 MIN (Grade) As min-grade,
 AVG (Grade) As avg-grade
FROM enrollments - table;

Output

Max-grade	min-grade	avg-grade
90	78	84,6

Question 10

SELECT course-id,
 COUNT(enrollment-id) As number of enrollment.
GROUP BY course-id

Output

Course-id	number of enrollments
101	1
102	1
103	1
104	1
105	1

Question 11

SELECT Department,
sum (salary) As total - salary,
sum (Bonus) As total - bonus
FROM Salaries - table.
GROUP BY Department;

Output

Department	total - salary	total - bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

Question 12

SELECT Department,
AVG (Salary) As avg-salary
FROM Salaries - table
GROUP BY Department
HAVING ~~AVG (Salary)~~ avg-salary > 55000.

Output.

Department	avg-salary
IT	61000
Finance	70000

Question 13

```
SELECT      employee - id,  
            name,  
            Salary,  
            Bonus,  
            Sum (salary + bonus) As total compensation  
FROM        Salaries - table  
WHERE       total - compensation > 60000;
```

Output:

Employee - id	Name	Salary	Bonus	total - compensation
1	Tom	60000	5000	65000
3	Jerry	70000	6000	76000
4	IT	62000	5500	67500

Question 14.

```
SELECT      Department,  
            Sum (Budget) As total - budget  
            Sum Avg (Budget) As avg - budget  
FROM        Projects - table  
GROUP BY    Department  
HAVING      avg - budget > 70000;
```

Output:

Department	total - budget	avg - budget.
IT	270000	135000
Finance	80000	80000

Question 15

SELECT *
FROM Projects-table
WHERE budget BETWEEN 50000 AND 120000 AND
department <> 'marketing';

Output

Project-id	project-name	department	budget
1	AI App	IT	120000
2	Payroll system	Finance	80000
5	HR Portal	HR	50000